JAKE MIRRA

MATHEMATICIAN · EDUCATOR · SOFTWARE ENGINEER

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SUMMARY

PhD mathematician and founding-team software engineer with fifteen years spanning research, classroom teaching, and production engineering. Built and scaled a real-estate operating platform from first pilot to institutional customers; applied mathematics and algorithms across human resources, accounting, education technology, and energy.

PROFESSIONAL EXPERIENCE

BASE POWER — Austin, TX — January 2026 – July 2026

Accounting and Financial Engineering

Financial and accounting engineering for a distributed residential energy company, working against energy infrastructure data at scale.

INDEPENDENT CONSULTING — 2024 – 2026

Mathematics, algorithms, and software consulting. Engagements in education technology and applied AI. Continued private tutoring in mathematics and physics.

TRILOGY — Education Technology — 2024 – 2025

Generative AI for educational content. Worked on a system to produce learning materials for high school students.

- Served as the translation layer between the pedagogy team and the engineering team, pre-reading education-team specifications to surface places where instructional intent would not survive contact with the implementation.
- Identified how structural patterns in the resolution of a mathematics problem determine its cognitive load and difficulty — work that made problem difficulty a predictable quantity rather than an editorial judgment.
- Characterized the recurring failure modes of large language models on generated mathematics, making model error predictable in advance rather than caught in review.
- Reviewed all generated output personally before handing it to the education team.

ENDER — Austin, TX — July 2019 – September 2024

Senior Software Engineer, March 2022 – September 2024 · Co-founder and Founding Engineer, July 2019 – March 2022

Property management and investment SaaS. Was writing its code before the company was formed, and stayed through institutional scale.

- Scaled the platform from mom-and-pop pilot customers to institutional investors operating tens of thousands of units.
- Maintained **100% uptime across two years** with no unplanned outages, under engineering-excellence policies I advocated for and implemented. Added platform health monitoring, invariant enforcement, and anomaly detection (SNS/SQS, OpsGenie).
- Architected, designed, and built the core accounting system backend in Java — the foundation of the product.
- Built automatic bank reconciliation by integrating Plaid with internal accounting data, plus a full UI suite for manual reconciliation, enabling larger operators to scale on the platform.
- Implemented secure bank linking and payments from tenant and vendor portals, integrated end to end with accounting.
- Designed and built APIs for tenant, vendor, investor, and property-manager portals, with a hierarchical, customizable permissions system.
- Built and deployed the data import pipeline that migrated new customers off legacy systems.
- Spearheaded the pilot with Ender's first major commercial real estate customer.
- Mentored and grew the development team; worked with sales, onboarding, product, and business teams to align technical strategy with company goals.

INDEED — Austin, TX — 2019

Technical Business Analyst, Search Quality

- Led design, implementation, and stakeholder management for an API bridge service for internal developers.
- Analysis showed the company was exposed to roughly **$100K/month in toll fraud** on a single third-party service; the bridge service closed that exposure and removed the need for developers to manually rotate API keys.
- Evaluated Java Spring, Flask, and Django before selecting the implementation stack; built the service in Django REST Framework with internal high-performance logging and indexing to hold latency down.

DULWICH COLLEGE INTERNATIONAL 2018 – 2019

Mathematics Curriculum Consultant

- Contract curriculum planning, design, and implementation on a team of over a dozen mathematicians, concurrent with the final year of doctoral work.
- Wrote Mathematica functions used by the whole team to produce consistent animations and visual content.
- All deliverables completed ahead of schedule.

UNIVERSITY OF PITTSBURGH — Department of Mathematics 2012 – 2018

Andrew Mellon Predoctoral Fellow · Lecturer · Teaching Assistant

- 10 semesters of teaching experience; 4 as lecturer of record in differential equations and linear algebra.
- Approximately 3,000 hours of tutoring, from high-school students through engineering undergraduates.
- Over a dozen technical talks during doctoral work, for both specialist and general audiences.

PRIVATE MATHEMATICS AND PHYSICS TUTOR 2014 – 2026

EDUCATION

PhD, Mathematics — University of Pittsburgh 2018

Thesis: *Hölder Continuous Mappings into Sub-Riemannian Manifolds.* Advisor: Piotr Hajłasz. Committee: Schikorra, Hales, DeBlois, Kaveh. 3.96 GPA across 72+ graduate credits in mathematics, including statistics, partial differential equations, analysis, and calculus of variations.

BS, Mathematics (minor in Computer Science) — University of Pittsburgh 2011 – 2012

Completed concurrently with entry into the graduate program.

Community College of Allegheny County 2010 – 2011

HONORS

Andrew Mellon Predoctoral Fellowship, University of Pittsburgh Dietrich School of Arts and Sciences (2013). Awarded by departmental nomination, by invitation only, to students of exceptional promise; carries stipend and tuition remission with no service requirement.

Montgomery M. Culver Award, Department of Mathematics, University of Pittsburgh (2012). Departmental recognition in mathematics, received during a first year at Pitt spent dual-enrolled between the undergraduate and graduate programs.

PUBLICATIONS

P. Hajłasz, **J. Mirra**, A. Schikorra, "Hölder continuous mappings, differential forms and the Heisenberg groups." *arXiv:2503.11506.* Develops distributional Jacobians and pullbacks of differential forms under Hölder continuous mappings, giving a simplified proof of a generalization of the Gromov non-embedding theorem and new results on the Hölder homotopy groups of the Heisenberg groups.

J. Mirra, "Hölder Continuous Mappings into Sub-Riemannian Manifolds." Doctoral dissertation, University of Pittsburgh, 2018. d-scholarship.pitt.edu/34970/

TECHNICAL SKILLS

MATHEMATICS	Geometric analysis, Sobolev and fractional Sobolev spaces, differential forms, linear algebra, Fourier analysis, PDE, calculus of variations, statistics, psychometrics and item response theory
ML / AI	LLM application development and agentic workflows; model failure-mode characterization and evaluation; TensorFlow, scikit-learn; hyperparameter search and regularization
CLOUD & DATA	Google Cloud Platform and AWS for data science at Ender and Base Power; BigQuery, SNS/SQS, relational schema design, ETL and migration pipelines
STACK	Java, Python, C++, JavaScript, SQL; Spring, Django REST Framework, Flask, Node.js, React; Mathematica
PRACTICE	Git and branch-scoped workflow, code review, twelve-factor configuration, deployment, performance monitoring